

Series : YXZW2



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कोड नं.
Code No. **55/2/3**

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.



भौतिक विज्ञान (सैद्धान्तिक)



PHYSICS (Theory)

निर्धारित समय : 3 घण्टे

Time allowed : 3 hours

अधिकतम अंक : 70

Maximum Marks : 70

नोट / NOTE

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(I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 23 हैं।

Please check that this question paper contains 23 printed pages.

(II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 33 प्रश्न हैं।

Please check that this question paper contains 33 questions.

(III) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।

Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.

(IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।

Please write down the serial number of the question in the answer-book at the given place before attempting it.

(V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।

15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.

55/2/3

729-3

Page 1 of 24

P.T.O.



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General Instructions :**Read the following instructions very carefully and follow them :**

- (i) *This question paper contains 33 questions. All questions are compulsory.*
- (ii) *This question paper is divided into five sections – Sections A, B, C, D and E.*
- (iii) *In Section A : Question numbers 1 to 16 are Multiple Choice type questions. Each question carries 1 mark.*
- (iv) *In Section B : Question numbers 17 to 21 are Very Short Answer type questions. Each question carries 2 marks.*
- (v) *In Section C : Question numbers 22 to 28 are Short Answer type questions. Each question carries 3 marks.*
- (vi) *In Section D : Question numbers 29 & 30 are case study-based questions. Each question carries 4 marks.*
- (vii) *In Section E : Question numbers 31 to 33 are Long Answer type questions. Each question carries 5 marks.*
- (viii) *There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the Sections except Section A.*
- (ix) *Kindly note that there is a separate question paper for Visually Impaired candidates.*
- (x) *Use of calculators is **not** allowed.*

You may use the following values of physical constants wherever necessary :

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\text{Mass of electron (m}_e\text{)} = 9.1 \times 10^{-31} \text{ kg.}$$

$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg.}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg.}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

$$\text{Boltzman's constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}$$



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SECTION – A

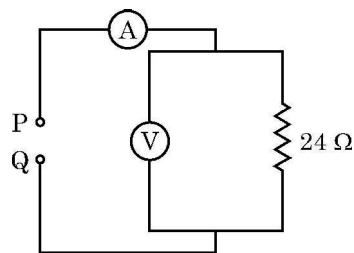
1. Consider two identical dipoles D_1 and D_2 . Charges $-q$ and q of dipole D_1 are located at $(0, 0)$ and $(a, 0)$ and that of dipole D_2 at $(0, a)$ and $(0, 2a)$ in x - y plane, respectively. The net dipole moment of the system is

1

- (A) $qa(\hat{i} + \hat{j})$ (B) $-qa(\hat{i} + \hat{j})$
(C) $qa(\hat{i} - \hat{j})$ (D) $-qa(\hat{i} - \hat{j})$

2. Which pair of readings of ideal voltmeter and ideal ammeter in the given circuit is possible when a suitable power source of $3\ \Omega$ internal resistance is connected between P and Q ?

1



- (A) 12.0 V, 2.0 A (B) 2.0 V, 0.5 A
(C) 6.0 V, 2.0 A (D) 12 V, 0.5 A

3. Which one of the following statements is correct ?

1

Electric field due to static charges is

- (A) conservative and field lines do not form closed loops.
(B) conservative and field lines form closed loops.
(C) non-conservative and field lines do not form closed loops.
(D) non-conservative and field lines form closed loops.

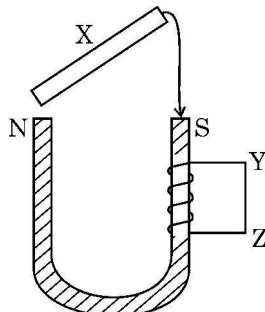
4. A material is pushed out when placed in a uniform magnetic field. The material is

1

- (A) non-magnetic (B) diamagnetic
(C) paramagnetic (D) ferromagnetic

5. A soft iron rod X is allowed to fall on the two poles of a U shaped permanent magnet as shown in figure. A coil is wrapped over one arm of the U shaped magnet.

1



During fall of the rod, the current in the coil will be

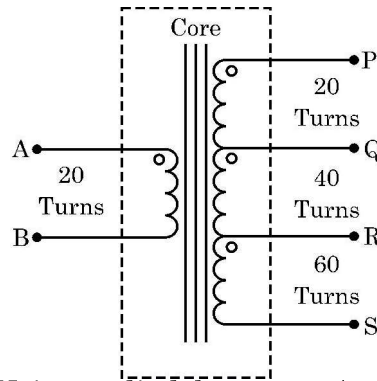
- (A) clockwise current (B) anticlockwise current
(C) alternating current (D) zero



6. A 1 cm straight segment of a conductor carrying 1 A current in x direction lies symmetrically at origin of Cartesian coordinate system. The magnetic field due to this segment at point (1m, 1m, 0) is

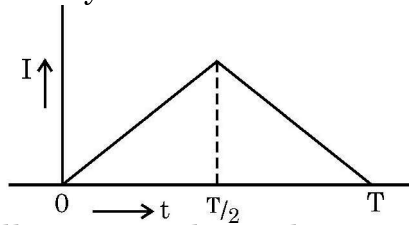
- (A) $1.0 \times 10^{-9} \hat{k}$ T (B) $-1.0 \times 10^{-9} \hat{k}$ T
(C) $\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k}$ T (D) $-\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k}$ T

7. The number of turns between different pairs of output terminals are shown for a step-up transformer.

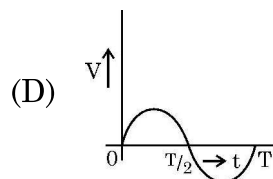
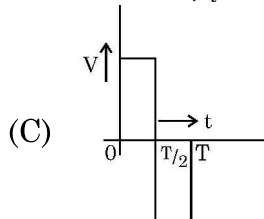
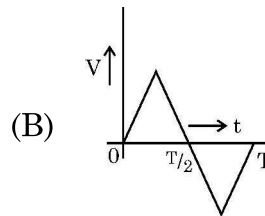
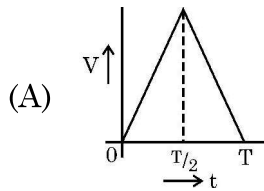


Input voltage of 20 V is applied between A and B. Between which two terminals will the output be 120 V ?

- (A) P and Q (B) Q and S
(C) P and R (D) P and S
8. The alternating current I in an inductor is observed to vary with time t as shown in the graph for a cycle.



Which one of the following graphs is the correct representation of wave form of voltage V with time t ?





9. The plane face of a planoconvex lens is silvered. The refractive index of material and radius of curvature of the curved surface of the lens are n and R respectively. This lens will behave as a concave mirror of focal length 1
- (A) $\frac{R}{n}$ (B) $\frac{R}{(n-1)}$
(C) nR (D) $\frac{R}{2(n-1)}$
10. When the resistance measured between p and n ends of a p-n junction diode is high, it can act as a/an – 1
- (A) resistor (B) inductor
(C) capacitor (D) switch
11. Atomic spectral emission lines of hydrogen atom are incident on a zinc surface. The lines which can emit photoelectrons from the surface are members of 1
- (A) Balmer series
(B) Paschen series
(C) Lyman series
(D) Neither Balmer, nor Paschen nor Lyman series
12. The energy of an electron in a hydrogen atom in ground state is -13.6 eV. Its energy in an orbit corresponding to quantum number n is -0.544 eV. The value of n is 1
- (A) 2 (B) 3
(C) 4 (D) 5

For Questions 13 to 16, two statements are given – one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below :

- (A) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) If both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) If Assertion (A) is true but Reason (R) is false.
(D) If both Assertion (A) and Reason (R) are false.

13. **Assertion (A)** : In an ideal step-down transformer, the electrical energy is not lost. 1
Reason (R) : In a step-down transformer, voltage decreases but the current increases.
14. **Assertion (A)** : Out of Infrared and radio waves, the radio waves show more diffraction effect. 1
Reason (R) : Radio waves have greater frequency than infrared waves.



15. **Assertion (A)** : In a semiconductor diode the thickness of depletion layer is not fixed. 1

Reason (R) : Thickness of depletion layer in a semiconductor device depends upon many factors such as biasing of the semiconductor.

16. **Assertion (A)** : In Bohr model of hydrogen atom, the angular momentum of an electron in n^{th} orbit is proportional to the square root of its orbit radius r_n . 1

Reason (R) : According to Bohr model, electron can jump to its nearest orbits only.

SECTION – B

17. The threshold voltage of a silicon diode is 0.7 V. It is operated at this point by connecting the diode in series with a battery of V volt and a resistor of 1000Ω . Find the value of V when the current drawn is 15 mA. 2

18. In a double slit experiment, it is observed that the angular width of one fringe formed on the screen is 0.2° . The wavelength of light used in the experiment is 500 nm. Calculate the separation of the two slits. 2

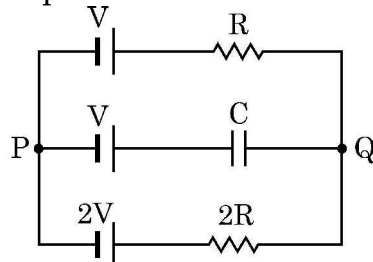
19. A light beam converges at a point O. In the path of this beam, a concave lens of focal length 15 cm is placed at a distance of 10 cm before point O. The beam now converges at a point O'. Find the magnitude and the direction of shift OO'. 2

20. The threshold wavelength of a metal is 450 nm. Calculate (i) the work function of the metal in eV and (ii) the maximum energy of the ejected photoelectrons in eV by incident radiation of 250 nm. 2

21. (a) Two wires of the same material and the same radius have their lengths in the ratio 2 : 3. They are connected in parallel to a battery which supplies a current of 15 A. Find the current through the wires. 2

OR

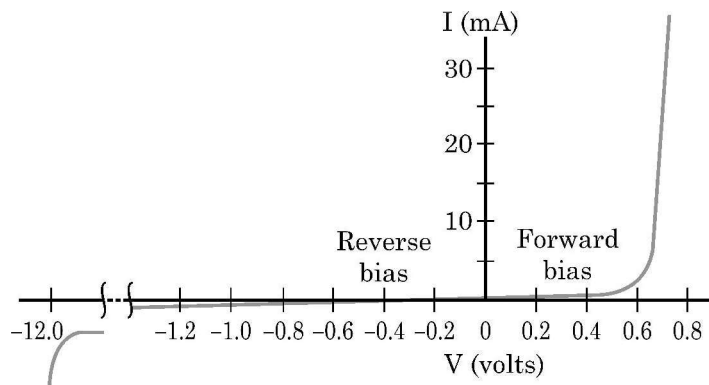
(b) In the circuit three ideal cells of e.m.f. V, V and 2V are connected to a resistor of resistance R, a capacitor of capacitance C and another resistor of resistance 2R as shown in figure. In the steady state find (i) the potential difference between P and Q and (ii) potential difference across capacitor C.





SECTION – C

22. (a) Define Electrical conductivity. Obtain the expression of electrical conductivity of a conductor in terms of number density and relaxation time of free electrons. 3
- (b) Explain qualitative change in resistivity of a conductor with temperature using expression obtained in (a).
23. (a) Show the variation of binding energy per nucleon with mass number. Write the significance of the binding energy curve. 3
- (b) Two nuclei with lower binding energy per nucleon form a nuclei with more binding energy per nucleon.
- (i) What type of nuclear reaction is it ?
- (ii) Whether the total mass of nuclei increases, decreases or remains unchanged ?
- (iii) Does the process require energy or produce energy ?
24. (a) ac voltage of frequency ω is applied across a series LCR circuit. Draw the phasor diagram and obtain the impedance of the circuit. 3
- (b) Discuss 'resonance' in a series LCR circuit and write the expression for resonant frequency.
25. (a) The amplitude of a light wave becomes n times. This results in intensity of the wave becoming m times. What is the relation between n and m ? 3
- (b) White light is incident on three identical surfaces – a black surface, a yellow surface and a white surface, one by one. For which surface, the pressure exerted on the surface by the incident light will be (i) maximum (ii) minimum ? Justify your answer.
26. (a) What are majority and minority charge carriers in an extrinsic semiconductor ? 3
- (b) A p-n junction is forward biased. Describe the movement of the charge carriers which produce current in it.
- (c) The graph shows the variation of current with voltage for a p-n junction diode.



Estimate the dynamic resistance of diode at $V = -0.6$ volt.



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27. (a) In a region of a uniform electric field $\vec{E}$, a negatively charged particle is moving with a constant velocity $\vec{v} = -v_0 \hat{i}$ near a long straight conductor coinciding with XX' axis and carrying current I towards $-X$ axis. The particle remains at a distance d from the conductor.
- (i) Draw diagram showing direction of electric and magnetic fields.
(ii) What are the various forces acting on the charged particle ?
(iii) Find the value of v_0 in terms of E , d and I .

3

OR

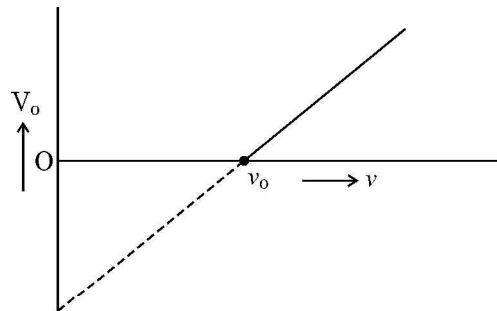
- (b) Two infinitely long conductors kept along XX' and YY' axes are carrying current I_1 and I_2 along $-X$ axis and $-Y$ axis respectively. Find the magnitude and direction of the net magnetic field produced at point $P(X, Y)$.
28. (a) When a parallel beam of light enters water surface obliquely at some angle, what is the effect on the width of the beam ?
(b) With the help of a ray diagram, show that a straw appears bent when it is partly dipped in water and explain it.
(c) Explain the transmission of optical signal through an optical fibre by a diagram.

3

SECTION - D

Question numbers 29 and 30 are case study based questions. Read the following paragraphs and answer the questions that follow.

29. When a photon of suitable frequency is incident on a metal surface, photoelectron is emitted from it. If the frequency is below a threshold frequency (ν_0) for the surface, no photoelectron is emitted. For a photon of frequency ν ($\nu > \nu_0$), the kinetic energy of the emitted photoelectrons is $h(\nu - \nu_0)$. The photocurrent can be stopped by applying a potential V_0 called 'stopping potential' on the anode. Thus maximum kinetic energy of photoelectrons $K_m = eV_0 = h(\nu - \nu_0)$. The experimental graph between V_0 and ν for a metal is shown in figure. This is a straight line of slope m . $4 \times 1 = 4$





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- (i) The straight line graphs obtained for two metals
(A) coincide each other.
(B) are parallel to each other.
(C) are not parallel to each other and cross at a point on ν -axis.
(D) are not parallel to each other and do not cross at a point on ν -axis.
- (ii) The value of Planck's constant for this metal is
(A) $\frac{e}{m}$ (B) $\frac{1}{me}$
(C) me (D) $\frac{m}{e}$
- (iii) The intercepts on ν -axis and V_0 -axis of the graph are respectively :
(A) $\nu_0, \frac{h\nu_0}{e}$ (B) $\nu_0, h\nu_0$
(C) $\frac{h\nu_0}{e}, \nu_0$ (D) $h\nu_0, \nu_0$

OR

- (iii) When the wavelength of a photon is doubled, how many times its wave number and frequency become, respectively ?
(A) $2, \frac{1}{2}$ (B) $\frac{1}{2}, \frac{1}{2}$
(C) $\frac{1}{2}, 2$ (D) $2, 2$
- (iv) The momentum of a photon is 5.0×10^{-29} kg. m/s. Ignoring relativistic effects (if any), the wavelength of the photon is
(A) $1.33 \mu\text{m}$ (B) $3.3 \mu\text{m}$
(C) $16.6 \mu\text{m}$ (D) $13.3 \mu\text{m}$

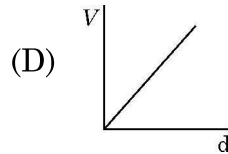
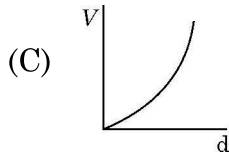
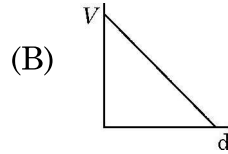
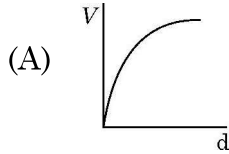
30. A parallel plate capacitor has two parallel plates which are separated by an insulating medium like air, mica, etc. When the plates are connected to the terminals of a battery, they get equal and opposite charges and an electric field is set up in between them. This electric field between the two plates depends upon the potential difference applied, the separation of the plates and nature of the medium between the plates. $4 \times 1 = 4$

- (i) The electric field between the plates of a parallel plate capacitor is E . Now the separation between the plates is doubled and simultaneously the applied potential difference between the plates is reduced to half of its initial value. The new value of the electric field between the plates will be :
(A) E (B) $2E$
(C) $\frac{E}{4}$ (D) $\frac{E}{2}$

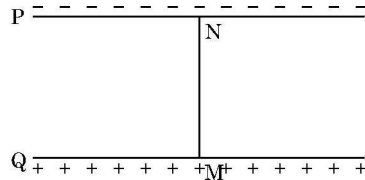


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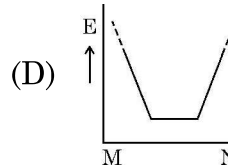
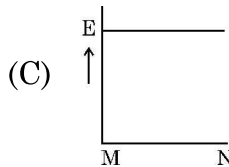
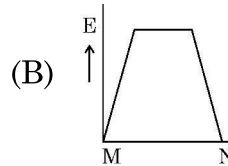
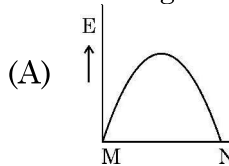
- (ii) A constant electric field is to be maintained between the two plates of a capacitor whose separation d changes with time. Which of the graphs correctly depict the potential difference (V) to be applied between the plates as a function of separation between the plates (d) to maintain the constant electric field ?



- (iii)



In the above figure P, Q are the two parallel plates of a capacitor. Plate Q is at positive potential with respect to plate P. MN is an imaginary line drawn perpendicular to the plates. Which of the graphs shows correctly the variations of the magnitude of electric field strength E along the line MN ?



- (iv) Three parallel plates are placed above each other with equal displacement $\vec{d}$ between neighbouring plates. The electric field between the first pair of the plates is $\vec{E}_1$ and the electric field between the second pair of the plates is $\vec{E}_2$. The potential difference between the third and the first plate is –

(A) $(\vec{E}_1 + \vec{E}_2) \cdot \vec{d}$

(B) $(\vec{E}_1 - \vec{E}_2) \cdot \vec{d}$

(C) $(\vec{E}_2 - \vec{E}_1) \cdot \vec{d}$

(D) $\frac{d(\vec{E}_1 + \vec{E}_2)}{2}$

OR



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- (iv) A material of dielectric constant K is filled in a parallel plate capacitor of capacitance C . The new value of its capacitance becomes
- (A) C (B) $\frac{C}{K}$
(C) CK (D) $C\left(1 + \frac{1}{K}\right)$

SECTION - E

31. (a) (i) What is the source of force acting on a current-carrying conductor placed in a magnetic field ? Obtain the expression for force acting between two long straight parallel conductors carrying steady currents and hence define 'ampere'. 5
- (ii) A point charge q is moving with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$. Find the work done by the magnetic force on the charge.
- (iii) Explain the necessary conditions in which the trajectory of a charged particle is helical in a uniform magnetic field.

OR

- (b) (i) A current carrying loop can be considered as a magnetic dipole placed along its axis. Explain.
- (ii) Obtain the relation for magnetic dipole moment $\vec{M}$ of current carrying coil. Give the direction of $\vec{M}$.
- (iii) A current carrying coil is placed in an external uniform magnetic field. The coil is free to turn in the magnetic field. What is the net force acting on the coil ? Obtain the orientation of the coil in stable equilibrium. Show that in this orientation the flux of the total field (field produced by the loop + external field) through the coil is maximum.
32. (a) (i) A thin pencil of length $(f/4)$ is placed coinciding with the principal axis of a mirror of focal length f . The image of the pencil is real and enlarged, just touches the pencil. Calculate the magnification produced by the mirror. 5
- (ii) A ray of light is incident on a refracting face AB of a prism ABC at an angle of 45° . The ray emerges from face AC and the angle of deviation is 15° . The angle of prism is 30° . Show that the emergent ray is normal to the face AC from which it emerges out. Find the refraction index of the material of the prism.

OR



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- (b) (i) Light consisting of two wavelengths 600 nm and 480 nm is used to obtain interference fringes in a double slit experiment. The screen is placed 1.0 m away from slits which are 1.0 nm apart.
- (1) Calculate the distance of the third bright fringe on the screen from the central maximum for wavelength 600 nm.
 - (2) Find the least distance from the central maximum where the bright fringes due to both the wavelengths coincide.
- (ii) (1) Draw the variation of intensity with angle of diffraction in single slit diffraction pattern. Write the expression for value of angle corresponding to zero intensity locations.
- (2) In what way diffraction of light waves differs from diffraction of sound waves ?
33. (a) (i) A small conducting sphere A of radius r charged to a potential V , is enclosed by a spherical conducting shell B of radius R . If A and B are connected by a thin wire, calculate the final potential on sphere A and shell B.
- (ii) Write two characteristics of equipotential surfaces. A uniform electric field of 50 NC^{-1} is set up in a region along $+x$ axis. If the potential at the origin $(0, 0)$ is 220 V, find the potential at a point $(4\text{m}, 3\text{m})$.

5

OR

- (b) (i) What is difference between an open surface and a closed surface ?
Draw elementary surface vector $d\vec{S}$ for a spherical surface S .
- (ii) Define electric flux through a surface. Give the significance of a Gaussian surface. A charge outside a Gaussian surface does not contribute to total electric flux through the surface. Why ?
- (iii) A small spherical shell S_1 has point charges $q_1 = -3 \mu\text{C}$, $q_2 = -2 \mu\text{C}$ and $q_3 = 9 \mu\text{C}$ inside it. This shell is enclosed by another big spherical shell S_2 . A point charge Q is placed in between the two surfaces S_1 and S_2 . If the electric flux through the surface S_2 is four times the flux through surface S_1 , find charge Q .